

Abhinav Gorantla

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Education

Arizona State University, *School of Computing and Augmented Intelligence* Jan 2026 – Present
Doctor of Philosophy in Computer Science

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.

Arizona State University, *School of Computing and Augmented Intelligence* Aug 2023 – Dec 2025
Master of Science in Computer Science, GPA: 4.0/4.0

- **Honors:** With Distinction.
- **Thesis:** *Speeding-up of the Non-Dominated Sorting Genetic Algorithm-II by Selective De-Correlation* 🔗
- **Advisor:** Dr. K. Selçuk Candan.

Vellore Institute of Technology, *School of Computer Science and Engineering* Jul 2019 – May 2023
Bachelor of Technology in Computer Science and Engineering

- **GPA:** 8.94/10.0

Publications

Causal Search for Skylines (CSS): Causally-Informed Selective Data De-Correlation

Pratanu Mandal, **Abhinav Gorantla**, K. Selçuk Candan, Maria Luisa Sapino
Accepted to ACM SIGMOD 2026.

Proposes a method for leveraging causal structure to selectively de-correlate data, improving efficiency of Skyline retrieval in relational databases.

Preprint: arxiv.org/abs/2603.14339 🔗

Code: github.com/EmitLab/Causal-Search-for-Skylines 🔗

CausalBench-ER: Causally-Informed Explanations and Recommendations for Reproducible Benchmarking

Ahmet Kapkic, Pratanu Mandal, **Abhinav Gorantla**, Shu Wan, Ertugrul Coban, Paras Sheth, Huan Liu, K. Selçuk Candan

In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), 2025.

Extends CausalBench with a recommendation engine that suggests appropriate causal analysis pipelines based on dataset, model and hyperparameter characteristics.

DOI: [10.1145/3746252.3761606](https://doi.org/10.1145/3746252.3761606) 🔗

Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning [Best Demo Award]

Ahmet Kapkic, Pratanu Mandal, Shu Wan, Paras Sheth, **Abhinav Gorantla**, Yoonhyuk Choi, Huan Liu, K. Selçuk Candan

In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), 2024.

An open benchmarking platform for causal learning and machine learning.

DOI: [10.1145/3627673.3679218](https://doi.org/10.1145/3627673.3679218) 🔗

Tutorials

CausalBench: Causal Learning Research Streamlined

Ahmet Kapkic, Pratanu Mandal, **Abhinav Gorantla**, Shu Wan, Ertugrul Coban, Paras Sheth, Huan Liu, K. Selçuk Candan

Tutorial at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025), 2025.

DOI: [10.1145/3711896.3737598](https://doi.org/10.1145/3711896.3737598) 

Research and Professional Experience

Graduate Research Assistant

EMIT Lab, Arizona State University

Tempe, AZ

August 2024 - (present)

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.
- Working on causally informed training methods for Prior Fitted Networks (PFNs).
- Developing an optimized algorithm for efficient Skyline retrieval in relational database systems by leveraging causal structure in the data.
- Conducting research on optimizing the selection process in multi-objective evolutionary algorithms, specifically Pareto-based methods like NSGA and SPEA.
- Collaborating with researchers at CASCADE Lab to maintain and improve causalbench.org, a platform dedicated to benchmark causal learning algorithms.
- Collaborated with researchers at CASCADE Lab in developing a novel causally informed recommendation system for causalbench.org.

Graduate Services Assistant

EMIT Lab, Arizona State University

Tempe, AZ

March 2024 - August 2024

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning.
- Supported CASCADE Lab researchers in developing the [causalbench-asu](https://causalbench-asu.github.io)  Python package and website, establishing an end-to-end benchmarking solution for the causal machine learning community.
- Served as a full stack developer on the [CausalBench](https://causalbench.org)  project, contributing to a comprehensive framework for benchmarking causal machine learning algorithms.

Software Development Engineer Intern

Webknot Technologies Pvt. Ltd.

Bengaluru, India

April 2022 - June 2023

- Revamped API endpoints within the Palette project, achieving a notable 30% reduction in response times.
- Engineered a custom plugin for Sisense BI software, enabling the seamless display of geojson data on a GeoJSON layer atop maps rendered via DeckGL.
- Fine-tuned data flow for the DeckGL plugin within Sisense by elevating the efficiency of JAQL queries, ensuring a smoother and more responsive user experience.

Projects

Research Publications Analysis tool

- Architected and developed a research publication analysis web application for ASU, integrating AWS S3, SageMaker, and SES. This tool leverages SCOPUS APIs to fetch research papers and perform text analysis of abstracts.
- Utilized LLM (GPT-4) for advanced feature extraction from scraped text.
- Optimized back-end system performance by reducing server response time by 80% and production costs by 40% through the strategic integration of AWS SageMaker.
- Tools Used: ReactJS, NodeJS, Python-FastAPI, RabbitMQ, MongoDB, AWS S3, AWS Sagemaker, OpenAI API.

Enhancing Diversity in the LLM Modulo Framework through Multi-Response Generation

- Developed the Diversified LLM Modulo framework to address looping and redundancy in the LLM Modulo

framework.

- Improved the performance of the LLM Modulo Framework on Planning tasks. Tested my framework on the Google Deepmind Natural Plan benchmark (paper) and achieved a performance improvement of 300% by increasing the diversity of LLM (Large Language Model) Responses.
- Tools Used: Python, OpenAI API.

Multimodal Image Retrieval System using Advanced Feature Analysis and Search Techniques

- Developed a Python-based image retrieval engine encompassing feature extraction from Caltech101 dataset images, latent semantics computation, clustering, and classification.
- Employed Locality Sensitive Hashing to index image features, optimizing nearest neighbor searches and ensuring scalability for expansive image datasets.
- Tools Used: Python.

Teaching Experience

ASU SCience and ENgineering Experience (SCENE)

Co-mentored a SCENE high school student on event detection in multivariate time-series data; supported experimental design and preparation for AZSEF.

Arizona State University
Spring 2026

Graduate Teaching Assistant

CSE 510 Database Management Systems Implementation

Arizona State University
Spring 2025

Graduate Teaching Assistant

CSE 515 Multimedia and Web Databases

Arizona State University
Fall 2024

Community Service

Student Volunteer

ACM Knowledge Discovery and Data Mining (KDD) 2025 Conference

Awards

University Graduate Fellowship, School of Computing and AI

Arizona State University

Skills

Programming: Python, C++, C, Java, TypeScript.

Machine Learning: PyTorch, NumPy, Pandas, Scikit-learn.

Databases Technologies: MongoDB, SQL, SQLAlchemy.

Cloud Technologies: AWS S3, AWS EC2, AWS Sagemaker, AWS Lambda, Google Firestore, Google Firebase Functions.

Web Technologies: NodeJS (family), ReactJS, FastAPI.

Other: Git, Tensorflow, RabbitMQ.

Leadership

Vice Chairperson

THEP Journalism Club, Vellore Institute of Technology

Vellore, India

- Established and sustained a robust MERN stack application as the cornerstone of the club's online newsletter platform.
- Pioneered the creation of a podcast for our club, achieving an impressive audience of nearly 800 listeners during its inaugural month.

Editor-in-Chief

Fifth Pillar NGO, Vellore Institute of Technology

Vellore, India

- Oversaw and directed a 50-member editorial team, responsible for the editing and publication of articles on

our blog website.

- Innovated by introducing new content formats like "Law Talks," resulting in a remarkable 50% expansion in the club's social media and community outreach.

⁰References available upon request.